Huawei Access Node HEB-ACC-40 - Alarm Handling for ALARM_7836 (Warning - Hardware/Environmental Fault)

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7836

Description

The failure of Node ID HEB-ACC-40 at Hebbal Edge, running on Huawei Access equipment, is attributed to a Hardware/Environmental Fault as indicated by the ALARM_7836 alarm code. This alarm is triggered when the node experiences ASIC memory parity errors, line card crashes, or a loss of power supply redundancy. The Huawei Access node is designed to operate in a redundant configuration, where the loss of a line card or power supply is automatically detected and rerouted to the standby component. However, in the event of a hardware or environmental fault, the node may not be able to recover, leading to a warning alarm being triggered. The underlying protocol mechanics are based on the IEEE 802.1D Spanning Tree Protocol (STP), which is used to manage the redundancy of the node's switching fabric. In this scenario, the STP may not be able to recover from the fault, leading to a loss of connectivity and the triggering of the ALARM_7836 alarm.

Pre-requisite Checks

- check: Verify that the Hebbal Edge site is accessible and that the Node ID HEB-ACC-40 is online., command: display current-configuration | include node HEB-ACC-40

- check: Verify that the redundant link status is stable and that there are no other alarms on the node., command: display alarm

- check: Verify that the node's power supply redundancy is healthy and that there are no other environmental issues., command: display environment

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Loss of power supply redundancy	Low

Diagnostic Commands & Expected Outputs

```
Command: display diagnose asic fault, Expected Output: ASIC fault information, Description: This command is used to display the ASIC fault information, including any parity errors that may have occurred.

Command: display diagnose card crash, Expected Output: Card crash information, Description: This command is used to display the card
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crash information, including any errors that may have occurred.

Command: display environment power supply, Expected Output: Power supply information, Description: This command is used to display the power supply information, including any redundancy issues that may have occurred.
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Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace asic slot , Description: This step involves replacing the faulty ASIC with a new one. The should be replaced with the actual slot number of the faulty ASIC.

Step 2: Step: Replace the faulty line card, Command: replace card slot , Description: This step involves replacing the faulty line card with a new one. The should be replaced with the actual slot number of the faulty line card.

Step 3: Step: Check and replace the power supply, Command: display environment power supply, Description: This step involves checking the power supply information to determine if the redundancy issue is related to the power supply. If the issue is related to the power supply, replace it with a new one.

Post-Check Verification

```
Command: display diagnose asic fault, Expected Output: No ASIC faults, Description: This command is used to verify that the ASIC fault has been resolved.

Command: display diagnose card crash, Expected Output: No card crashes, Description: This command is used to verify that the card crash has been resolved.

Command: display environment power supply, Expected Output: Power supply redundancy is healthy, Description: This command is used to verify that the power supply redundancy issue has been resolved.
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Roll-back Procedures

- Step: Restore the original ASIC, Command: replace asic slot original, Description: This step involves restoring the original ASIC. The should be replaced with the actual slot number of the original ASIC.

- Step: Restore the original line card, Command: replace card slot original, Description: This step involves restoring the original line card. The should be replaced with the actual slot number of the original line card.

- Step: Restore the original power supply, Command: replace power supply slot original, Description: This step involves restoring the original power supply. The should be replaced with the actual slot number of the original power supply.